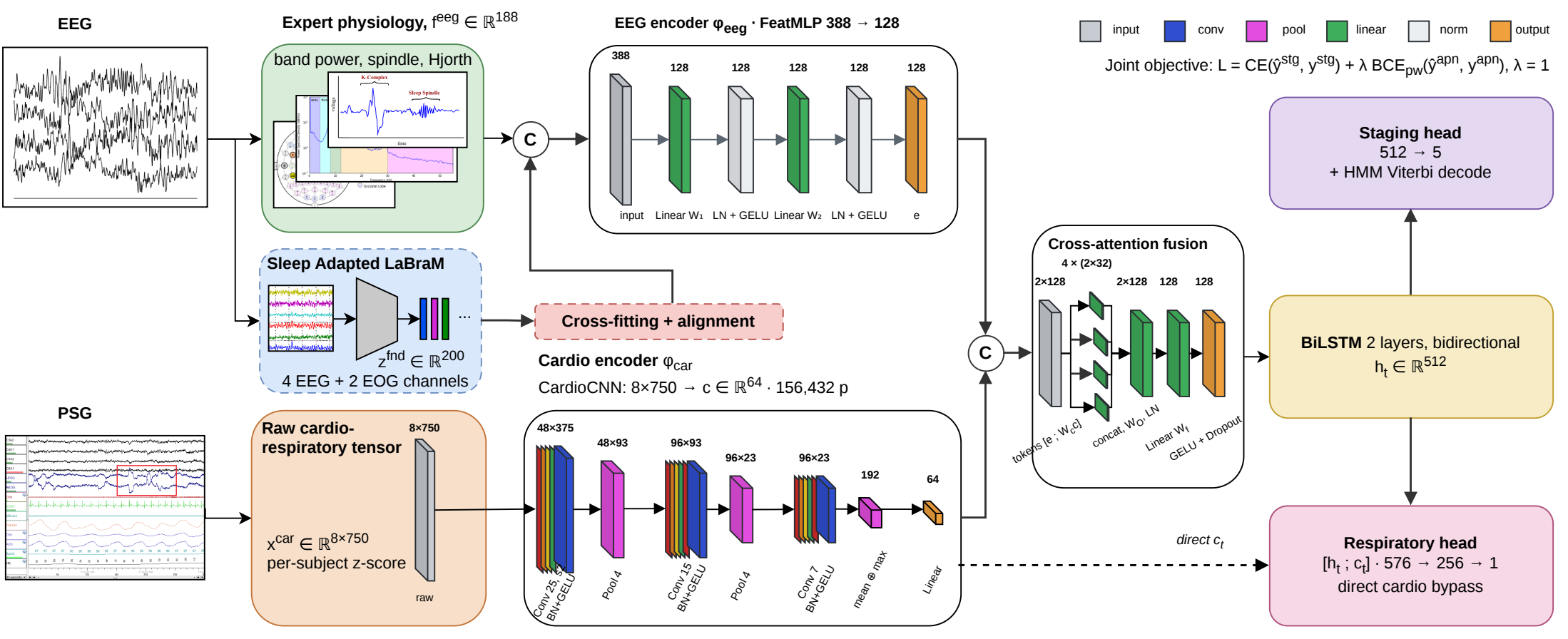

EEG
Expert physiology, $f^{eeg} \in \mathbb{R}^{188}$
band power, spindle, Hjorth
K-Complex
Sleep Spindle
EEG encoder φ_{eeg} · FeatMLP 388 → 128
388
128
128
128
128
128
input
Linear W_1
LN + GELU
Linear W_2
LN + GELU
e
input
conv
pool
linear
norm
output
Joint objective: $L = CE(\hat{y}^{stg}, y^{stg}) + \lambda\, BCE_{pw}(\hat{y}^{apn}, y^{apn})$, $\lambda = 1$
Staging head
512 → 5
+ HMM Viterbi decode
Sleep Adapted LaBraM
$z^{fnd} \in \mathbb{R}^{200}$
4 EEG + 2 EOG channels
Cross-fitting + alignment
Cardio encoder φ_{car}
CardioCNN: 8×750 → $c \in \mathbb{R}^{64}$ · 156,432 p
Cross-attention fusion
4 × (2×32)
2×128
2×128
128
128
tokens $[e ; W_c c]$
concat, W_O, LN
Linear W_f
GELU + Dropout
BiLSTM 2 layers, bidirectional
$h_t \in \mathbb{R}^{512}$
PSG
Raw cardio-respiratory tensor
8×750
$x^{car} \in \mathbb{R}^{8\times750}$
per-subject z-score
raw
48×375
48×93
96×93
96×23
96×23
192
64
Conv 25, 5× BN+GELU
Pool 4
Conv 15 BN+GELU
Pool 4
Conv 7 BN+GELU
mean ⊕ max
Linear
direct c_t
Respiratory head
$[h_t ; c_t] \cdot 576 \rightarrow 256 \rightarrow 1$
direct cardio bypass